



Adobe Comments on the National Telecommunications and Information Administration's AI Accountability Policy Request for Comment

Adobe Inc. (Adobe) appreciates the opportunity to submit comments in response to the National Telecommunications and Information Administration (NTIA) AI Accountability Policy Request for Comment.¹ We recognize the importance of building strong accountability mechanisms to ensure artificial intelligence (AI) systems are designed, developed, and deployed in a trustworthy and responsible manner, and we look forward to opportunities to engage further with the NTIA on this important topic.

Executive Summary

Adobe recommends a flexible approach to trustworthy AI that incentivizes AI accountability mechanisms guided by both government-developed standards and general principles of accountability, responsibility, and transparency.² This response highlights three themes:

- **Risk-based approach.** Adobe supports a risk-based approach to AI in line with the National Institute of Standards and Technology's (NIST) *AI Risk Management Framework* and emerging AI laws around the world, such as the European Union's original draft *Artificial Intelligence Act* (AIA). We encourage the government to leverage existing consumer protection and civil rights laws that mitigate harms while being technology neutral.
- **Content Credentials.** Adobe urges the government to promote transparency in digital content by driving the implementation of Content Credentials—open-sourced technology enabling creators to attach digital provenance information to digital content so that consumers can see the origins and edit history of content online. The 1,000+ member [Content Authenticity Initiative](#) (CAI) and its underlying standards body, the [Coalition for Content Provenance and Authenticity](#) (C2PA), are working to drive adoption and implementation of this free, open-source technology across industries and platforms. Adobe strongly supports the government's leveraging of the C2PA standard to:
 - support greater transparency and trust in our online ecosystem by helping users understand the origins and edits of digital content;
 - enable automatic disclosure of AI-generated content;
 - encourage industry adoption of a Do Not Train standard for AI systems; and
 - endorse this industry-developed standard for other organizations and for use by the government.
- **Standardized benchmarks and test datasets.** Adobe recommends a specific role for the government in publishing standardized datasets to enable the testing of bias in AI systems.

¹ National Telecommunications and Information Administration, [AI Accountability Policy Request for Comment](#), Docket Number 230407-0093. (NTIA Consultation Paper).

² NTIA Consultation Paper at q. 1.

I. Background

At Adobe, our mission is to empower everyone to deliver the world's best digital experiences. Since our founding in December 1982, we have continued to pioneer transformative technologies that allow our customers—who range from emerging artists to global brands—to channel their imaginations, unleash their creativity, and power their businesses.

Adobe's business is comprised of three cloud-based solutions: Creative Cloud, Document Cloud, and Experience Cloud. In Adobe Creative Cloud, our tools like Photoshop, Illustrator, and Premiere Pro empower millions of creators around the world to create compelling assets like photos, videos, and illustrations, while our online and mobile design app Adobe Express allows people of all skills to easily create standout social graphics, videos, and web pages. In Document Cloud, our products provide individuals to small and medium businesses to large enterprises with tools to support all of their digital document needs. Lastly, our Experience Cloud business supports businesses with their websites and e-commerce experiences.

Across all of our products, Adobe is building on a decade-long legacy of AI innovation by leveraging the power of AI to deliver hundreds of intelligent capabilities. In Creative Cloud, AI powers many creative functions including advanced image editing features in Photoshop, making it easier for everyone to tell their story with simpler and more intuitive tools.

We recently launched Adobe Firefly, our new family of creative generative AI models designed to be both creator-focused and safe for commercial use.³ With Firefly, users can channel their creativity in ways they never imagined possible by simply typing in a prompt and generating images in seconds.

In Document Cloud, AI powers features like Liquid Mode, which automatically converts a PDF layout to different formats to fit different screens so it can be easily read on tablets, mobile devices and more. As part of our Digital Experience offerings, Adobe's customers can use AI-driven features to deliver relevant and meaningful insights and personalized digital experiences to the millions of visitors on their websites.

We believe that AI done right will amplify human creativity and capabilities to new levels with deeper insights, accelerated task performance, and improved decision-making ability. As we continue to harness the power of AI, we are committed to developing and deploying AI in line with our AI Ethics principles of accountability, responsibility, and transparency, while considering its broader impact on society.⁴

We have been active in this space and have provided our perspective to the White House Office of Science and Technology Policy as they developed the *AI Bill of Rights* and have testified before the European Parliament's Special Committee on AI in the Digital Age (AIDA), among other policy initiatives.

³ TechCrunch, "[Adobe's thoughts on the ethics of AI-generated images \(and paying its contributors for them\)](#)," Frederic Lardinois (March 22, 2023).

⁴ Adobe Blog, "[Seeing is believing: It's time to restore trust in online media](#)," Dana Rao (May 15, 2023).

II. Adobe's Comments

Globally and within the United States, governments are establishing rules and standards to promote AI assurance.⁵ When standards are targeted, contextual, and harmonized across jurisdictions, they can be very valuable in advancing trustworthy AI and reducing harms related to AI systems.

At Adobe, we have adopted an integrated approach to data governance and AI risk management by augmenting existing cyber incident response plans, data governance safeguards, and security assessments to account for risks associated with AI. We recommend that any government guidance on AI risk management leverage existing data protection principles, standards, and best practices wherever possible.⁶

A. Adobe Supports AI Accountability and Transparency Mechanisms

It is crucial not to overregulate at these early stages of AI research and development, but rather to enable businesses to invest and innovate responsibly. Adobe accepts NIST's characterization of "trustworthy AI" as "valid and reliable, safe, secure and resilient, accountable and transparent, explainable and interpretable, privacy-enhanced, and fair with harmful bias managed".⁷ In Adobe's experience, companies, both independently and in collaboration with others, are best placed to establish AI accountability mechanisms that can action these characteristics of trustworthy AI.

i. **Adobe's Proactive Internal AI Accountability Mechanisms**

Adobe agrees that "[t]he most useful audits and assessments of these [AI] systems, therefore, should extend beyond the technical to broader questions about governance and purpose".⁸ We lead by example. Beginning in 2019, we proactively and voluntarily developed a layered, multi-disciplinary process for responsible AI.

First, we established AI Ethics principles of accountability, responsibility, and transparency to guide the development of our AI-powered solutions.⁹

- **Accountability** means we take ownership of the outcomes of our AI-assisted tools. We have processes and resources dedicated to receiving and responding to concerns about our AI products and taking corrective action as appropriate. Accountability also requires testing for and anticipating potential harms, taking pre-emptive steps to mitigate such harms, and maintaining systems to respond to unanticipated harmful outcomes.

⁵ Adobe accepts the description of "AI assurance" referenced in the NTIA Consultation Paper: "...an AI system is legal, effective, ethical, safe and otherwise trustworthy – a function also known as providing AI assurance".

⁶ NTIA Consultation Paper at qq. 11, 27. We are seeing the use of AI addendums, similar in nature to the Data Protection Addendum in the EU's General Data Protection Regulation (GDPR), to set out the intended purpose and use of the AI system, IP and data rights, ongoing monitoring requirements, and other accountability requirements. This is one effective mechanism at ensuring a shared understanding of how the AI system is to be used and monitored, and by whom.

⁷ NTIA Consultation Paper at p. 22434.

⁸ NTIA Consultation Paper at p. 22436.

⁹ <https://www.adobe.com/about-adobe/aiethics.html>

- **Responsibility** means we approach designing and maintaining our AI technology with thoughtful evaluation and careful consideration of the impact and consequences of its deployment. We ensure that we design for inclusiveness and assess the impact of potentially unfair, discriminatory, or inaccurate results, which might perpetuate harmful biases and stereotypes. We understand that special care must be taken to address bias if a product or service will have a significant impact on an individual's life, such as with employment, housing, credit, and health.
- **Transparency** means we are open about, and explain, our use of AI to our customers so they have a clear understanding of our AI systems and their applications. We want our customers to understand how Adobe uses AI, the value AI-assisted tools bring to them, and what controls and preferences they have available when they engage with and utilize Adobe's AI-enhanced tools and services.

Guided by these principles, we have created standardized processes from design to development to deployment, including training, testing, and a review process overseen by a diverse AI Ethics Review Board.

Second, we have set up an AI Ethics Committee and an AI Ethics Review Board to action ethical concerns with new features and technologies.

- Our **AI Ethics Committee** is a diverse, cross-functional team with members from across Adobe that is responsible for helping to ensure the AI Ethics principles are understood and incorporated across our development teams.
- Our **AI Ethics Review Board** is tasked with the review of AI-powered features and products before their release. Like our AI Ethics Committee, the AI Ethics Review Board is a cross-functional group of Adobe employees with diverse professional, gender, and ethnic backgrounds as we believe diversity is critical to identifying potential issues that a non-diverse team might not see.

In addition, we have an AI Ethics team that serves two main functions: (a) it advises our product and development teams on ethical considerations for our AI-powered solutions; and (b) it thinks expansively about ethics across various types of tools and technologies, both developed by Adobe for our customers and used internally by our teams.

Third, we have embedded an **AI Ethics Assessment** in our processes. The AI Ethics Assessment is a multi-part review that is designed to assess the ethical impact of a product or feature. If an initial assessment shows no major ethical impact and the feature meets our ethical standards, the feature is approved. AI features that have a higher potential ethical impact go through a more rigorous testing process, including a review by the AI Ethics Review Board. The AI Ethics Review Board will either approve the release; give a qualified approval with instructions to test or fix minimal issues; or not approve due to significant testing or remediation required. Where approval is denied, the product team is required to take on additional work and re-present to the AI Ethics Review Board.

These internal accountability mechanisms have been effective at upholding our AI Ethics principles. For example, our AI Ethics Review Board worked with Adobe development teams on a new suite of features in Photoshop called Neural Filters, which let users add non-destructive, generative filters to create things that were not previously in images using AI. Neural Filters allow for easy adjustment of someone's age or

expression, or can turn a black and white photo to color, by simply adjusting a slider. The Neural Filters development team used our AI Ethics Assessment to assess the potential ethical impact of the filters. When evaluating Neural Filters, one Review Board member flagged that one of the filters did not always properly model the hairstyle of a particular group. Based on this feedback, the Neural Filters development team updated the dataset to enable retraining of the AI model to address the issue before the feature went live.

Another example is when a member of the Review Board flagged early in the development process that a skin-smoothing feature in Neural Filters was removing tattoos from people's faces. Tattoos are culturally significant for certain communities. By removing a person's tattoo, the AI model could have been unintentionally changing someone's identity and conforming them to certain beauty norms. Since our Review Board is comprised of a diverse group of employees all reviewing the AI from their own personal perspective, we identified this issue early and shared this important feedback with our product and engineering teams. The teams made improvements to the model and updated the AI dataset before Neural Filters was released.

Finally, we promote and rely on community feedback in developing our AI products. For example, we launched our AI art generator Firefly as a beta in multiple stages to seek input from our employees, customers, and our broader creative community. Firefly has a feedback mechanism built in so that users can easily report if a feature produces a result they perceive as, for example, biased or inaccurate. We monitor this feedback data to identify any undesirable outcomes and allow our product and engineering teams to address them appropriately. In our view, this constant feedback loop with our user community is one important way to help ensure our tools minimize harm and uphold our values as a company.

Adobe supports the government's role in continuing to work with interested stakeholders, including the private sector, to set out technical and governance best practices for implementing trustworthy AI. We also recognize and endorse government initiatives that seek to use appropriate levers, including in its own relationship with vendors, to incentivize voluntary adoption of AI accountability mechanisms.¹⁰

ii. Adobe Leads Digital Provenance for AI Transparency Mechanisms

Adobe works with interested parties to develop practical and effective processes to achieve accountable AI. Adobe leads the [Content Authenticity Initiative](#) (CAI)—a global coalition working to increase transparency in digital content through open industry standards. Founded in 2019, the CAI now has more than 1,000 members from media outlets to camera manufacturers to generative AI developers and more. The CAI promotes the use of technology to allow creators to attach metadata, called Content Credentials, to a piece of content, including information such as name, date, and what tools were used to create it. Content Credentials remain associated with the content wherever it is used, published, or stored, and can help content viewers make more informed decisions about whether to trust it. This level of transparency also enables proper attribution for creators and helps fight misinformation.

In the context of generative AI, Content Credentials can indicate whether a piece of content was human-created, AI-edited, or AI-generated. Adobe automatically attaches Content Credentials to images

¹⁰ NTIA Consultation Paper at q. 32

generated in our generative AI tool, Firefly, to indicate that AI was used.¹¹

Finally, creators can use Content Credentials to attach a “Do Not Train” tag to their content if they don’t want their work to be used to train an AI model. With broad industry adoption, Content Credentials will help prevent web-crawlers from using works with “Do Not Train” credentials as part of a dataset.

Content Credentials are a free, open-source technology, based on an open standard created by the [Coalition for Content Provenance and Authenticity](#) (C2PA), which Adobe co-founded. Anyone can incorporate them into their own tools and products and Adobe is committed to working with our cross-industry partners to drive widespread implementation of the C2PA standard.

iii. Adobe Supports Internal Assessments as Effective Mechanisms for Real-Time Harm Identification and Prevention

In our experience, the companies that develop AI solutions are best placed to identify and address potential AI-related risks given their familiarity with the solutions and the intended use cases. Unlike dynamic internal assessment processes, external audits are typically static and point-in-time and do not account for the adaptable nature of AI. Therefore, Adobe cautions against the growing interest in mandating external assessments or audits of AI solutions. External audits typically promote “checkbox compliance” rather than a culture of AI ethics by design. External audits are also often expensive and cumbersome, thereby entrenching AI development within large technology companies.¹²

At Adobe, we emphasize internal and external beta testing of our AI models with trusted third parties to help stress test our models before commercial release. In addition to providing valuable feedback, this practice also aligns with our commitment to soliciting and receiving diverse perspectives on potential impacts of our AI-powered solutions.

Adobe strongly recommends that government work with the private sector to develop clear and consistent guidance and standards for internal AI audits and assessments. In so doing, the government can equip companies with methodologies to identify and mitigate AI-related harms. Adobe supports the government providing sector-specific guidance to ensure appropriate oversight without ineffective or burdensome external audit requirements.¹³

We further support leveraging existing civil rights laws and consumer protection laws that protect important rights and liberties in a manner that is technology-agnostic. These existing mechanisms should be favored over new, fragmented laws that risk overregulating AI innovations.

¹¹ N.Y. Times, [“A.I. Photoshopping Is About to Get Very Easy. Maybe Too Easy,”](#) Farhad Manjoo (May 23, 2023).

¹² NTIA Consultation Paper at q. 28.

¹³ NTIA Consultation Paper at q. 30.

B. Adobe Supports a Harmonized, Risk-based Approach Applied to the Lifecycle of the AI System

In 2021, there were more than 130 AI-focused bills proposed or passed in the US alone.¹⁴ Globally, the European Union, China, Brazil, Canada, Singapore, among others including multilateral organizations such as the Organization for Economic Cooperation and Development (OECD), are standing up policy or law on AI. Too much fragmentation makes it more difficult for global companies such as Adobe to compete and comply.¹⁵ Adobe favors aligning AI standards throughout the US and globally. Without compromising American values and culture, Adobe supports the harmonization of leading AI governance frameworks and laws as key to a pro-innovation approach to trustworthy AI.

To this end, Adobe strongly recommends that the government endorse a risk-based approach to AI accountability,¹⁶ which is present in existing guidance such as NIST's AI Risk Management Framework (RMF). The AI RMF has been developed through significant public consultation and "is intended to be flexible and to augment existing risk practices which should align with applicable laws, regulations, and norms".¹⁷ The risk-based approach is important to focusing regulatory efforts on high-risk AI without slowing down the pace of innovation around AI that poses little or no risk (e.g. AI used to recommend fonts based on a document template). Adobe is also supportive of the risk-based approach originally outlined in the EU's AI Act and recommends a similar approach to harmonize a global framework for effectively combatting the potential harmful effects of AI.

Adobe agrees that AI accountability mechanisms should be scoped to cover all stages of the AI lifecycle, ranging from ideation through design, development, deployment, and ongoing monitoring. AI accountability mechanisms can be used to effectively deal with systemic and/or collective risks of harm.¹⁸ This approach is adopted by our own internal AI accountability process and proves effective at early-detection and mitigation of AI-related harms.

C. Adobe Urges the Government to Embrace Content Credentials Requirements for Digital Content

As noted above, the CAI and its underlying standards body, the C2PA, establish processes and standards to promote trust in AI-powered content solutions. The CAI and C2PA showcase how industry collaborations can promote safety, reliability, trust, accountability, and transparency of AI outputs. They are examples of creative, stakeholder-led initiatives that help to protect public trust in digital content.

Adobe strongly supports the government's leveraging of the C2PA standard to:

- support greater transparency and trust in our online ecosystem by helping users understand the origins and edits of digital content;
- enable automatic disclosure of AI-generated content;
- encourage industry adoption of a Do Not Train standard for AI systems; and
- endorse this industry-developed standard for other organizations¹⁹ and for use by the

¹⁴ [Center for Data Innovation, U.S. AI Policy Report Card, July 27, 2022.](#)

¹⁵ NTIA Consultation Paper at q. 34.

¹⁶ NTIA Consultation Paper at q. 7.

¹⁷ [National Institute of Standards and Technology, Artificial Intelligence Risk Management Framework \(AI RMF 1.0\), January 2023](#), p. 7.

¹⁸ NTIA Consultation Paper at qq. 4, 15.

¹⁹ NTIA Consultation Paper at q. 9, 31.

government.²⁰

Encouraging the implementation of the C2PA standard would allow consumers to use Content Credentials to make more informed decisions about whether to trust content they are viewing online. This level of transparency also enables proper attribution for creators and helps fight misinformation.

In the context of generative AI, Content Credentials can indicate whether a piece of content was human-created, AI-edited, or AI-generated. Creators can use Content Credentials to attach a “Do Not Train” tag to their content if they do not want their work to be used to train an AI model. With broad industry adoption, Content Credentials will help prevent web-crawlers from using works with “Do Not Train” credentials as part of a dataset.

Adobe urges the government to incorporate digital content provenance and authenticity in any of its guidance on responsible AI as effective tools for protecting consumers and creators alike. Specifically, platforms that generate content with AI should be required to include and preserve Content Credentials for content on their platforms. We believe the transparency obligations for AI systems as originally outlined in the European Union’s AI Act is an important step in bringing more trust to digital content. By requiring AI-generating systems to implement technologies that enable disclosure of AI-generated content, the EU AI Act helps consumers understand and make more informed decisions about content they are consuming. We encourage NTIA to align its approach with the EU AI Act to foster a harmonized global framework toward ensuring a trustworthy digital space in the age of AI.

D. Adobe Strongly Recommends that the Government Curate Standardized Test Datasets

Adobe acknowledges the Consultation Paper’s specific request for feedback on policy proposals related to “making test data available for use.”²¹ We see that many of the draft regulations around AI development are aimed at mitigating harmful bias. Mitigating harmful bias is important to us at Adobe. It is the reason we established our comprehensive AI Ethics program that includes testing, training, and a review with our AI Ethics Board.

Government-based standardized test datasets would enable companies to run their AI models against these datasets to make sure their models meet certain performance and ethical standards. If a company’s models perform within an agreed-upon “normal range” in a distribution of outcomes, that company could get a certificate of compliance.²² The standardized datasets would need to work across various verticals, industries, markets, and use cases. There is an opportunity for the private sector to partner with governments to establish appropriate datasets in each case. We think this initiative would help create an environment where companies do not have to make a tradeoff between innovation and ethics.²³

²⁰ NTIA Consultation Paper at q. 30.

²¹ NTIA Consultation Paper at, p. 22438.

²² And if external audits were to be required, common standards—including test datasets—developed by the government will promote consistent and meaningful assessments by external auditors.

²³ NTIA Consultation Paper at q. 22.

III. Conclusion

Adobe appreciates the opportunity to provide comments on NTIA's Consultation Paper. We appreciate that NTIA has taken important steps to consult with interested stakeholders to leverage existing AI accountability mechanisms, minimize barriers to AI development and adoption, and identify opportunities to harmonize such standards across the United States and globally. We agree that AI accountability standards and guidance are needed. We encourage the use of industry-developed initiatives and standards such as the CAI and its underlying standards body, the C2PA, as examples of best practices to maintain authenticity and trust while mitigating harms caused by some AI solutions.

We look forward to continuing the discussion with NTIA acting as a hub for the broader community of interest, and we are eager to engage collaboratively on these important issues.